

TAHIA ANAN DHIRA

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PROFILE

Computational social scientist (MSc Applied Social Data Science, LSE, Distinction) working across social network analysis, quantitative text analysis, and applied econometrics. Currently transitioning from a teaching-intensive faculty appointment to a full-time research position ahead of doctoral study, with four active research manuscripts (2026) built on reproducible Python/R pipelines and a peer-reviewed record including PLOS ONE and SSM – Health Systems.

COMPUTATIONAL & QUANTITATIVE SKILLS

- Programming: Python (pandas, NumPy, NetworkX, Matplotlib, scikit-learn) · R (igraph, statnet, ergm, tergm, stm, networkDynamic, quanteda) · STATA
- Network analysis: ERGM, STERGM, dynamic and weighted networks, centrality, community detection, core-periphery analysis, block modelling, cluster analysis
- Econometrics: panel-data and time-series methods (fixed/random effects, PCSE, FGLS, GMM, ARDL) selected according to data structure — integration order, cross-sectional dependence; factor analysis (EFA/CFA), principal component analysis, structural equation modelling
- Machine learning & NLP: sentiment analysis (VADER), structural topic modelling (STM), applied machine learning; additional training in word embeddings, LDA, Wordfish, and transformer-based text classification (BERT)
- Research workflow: Git/GitHub; Gemini API; reproducible pipelines; data visualization; LaTeX
- Data collection: Qualtrics; YouTube Data API v3

EDUCATION

MSc in Applied Social Data Science (Distinction) — London School of Economics and Political Science (2023)

Coursework: Social Network Analysis; Quantitative Text Analysis; Applied Machine Learning; Managing and Visualizing Data; Research Design

MSS in Economics (CGPA 3.88/4.00) — University of Dhaka (2017)

BSS in Economics (CGPA 3.94/4.00) — University of Dhaka (2016)

CURRENT RESEARCH MANUSCRIPTS

Rethinking the Income–Emissions Relationship: Evidence from Trade-Embodied CO₂ Emissions Networks (*Submitted, 2026*)

- Built reproducible computational workflows integrating large-scale international trade, emissions, and macroeconomic datasets
- Constructed weighted global trade networks and ran longitudinal network analyses of embodied-carbon trade structure
- Combined network analysis with panel econometrics to examine how development and network position shape domestic and trade-embodied CO₂ emissions

Global Migration Network and the Geography of Development (*Working paper, 2026*)

- Built a 23-year longitudinal panel of directed, weighted bilateral migration networks
- Developed reproducible R workflows for network construction, visualization, and longitudinal analysis
- Contributed to network model development, core-periphery detection, and manuscript preparation

Product Reproducibility and International Trade Patterns: Evidence from Cultural Goods (*Under review, 2026*)

- Constructed computational pipelines harmonizing and processing trade data across 190 countries
- Combined structural gravity modelling (PPML) with dynamic network modelling (STERGM) to examine trade-relationship formation and persistence
- Corresponding author; built reproducible workflows across Python, R, and Git

Gendered Cultural Hierarchies in Social Media Audience Discourse: Evidence from Bangladeshi Artists (*Working paper, 2026*)

- Built scalable Python pipelines (YouTube Data API v3) to collect and process over 8 million multilingual comments
- Applied sentiment analysis (VADER) and structural topic modelling (quanteda/STM) to audience discourse across artists, genders, and artistic categories
- Designed a human-in-the-loop classification workflow using the Gemini API, with manual validation across ~48,000 videos

PEER-REVIEWED PUBLICATIONS

- Huque, R., Abdullah, S. M., **Dhira, T. A.**, et al. (2023). Determinants of client satisfaction with health insurance: Evidence using a mixed-method approach from Bangladesh. *SSM – Health Systems*, 3.
- Rahman, M. A., **Dhira, T. A.**, et al. (2022). PHQ-9 validation in Bangladesh. *PLOS ONE*.
- **Dhira, T. A.**, Rahman, M. A., et al. (2021). GAD-7 validation in Bangladesh. *PLOS ONE*.
- Mehareen, J., Rahman, M. A., **Dhira, T. A.**, et al. (2021). Mental health during COVID-19. *Journal of Affective Disorders Reports*.

EARLIER RESEARCH EXPERIENCE

Capstone Project, MSc ASDS, LSE — Friendship Network Formation: Pro-sociality and Personality Type (*2023*)

- Designed and implemented a behavioural classroom experiment on homophily and friendship network formation
- Estimated Exponential Random Graph Models (ERGMs) to identify structural and attribute-based drivers of network formation

Coursework project, ST445, LSE — Topic-Based Content and Sentiment Analysis of COVID-19 Vaccine Discourse on Reddit (*2022–2023*)

- Contributed to manuscript preparation and network visualization

Mental Health of University Students Before and After Online Education (UGC-funded) (*2020–2021*)

- Contributed to design and implementation of a longitudinal survey; coordinated data collection and survey-instrument validation

Willingness to Pay for Treating Mental vs. Physical Illness (BMRC-funded) (*2020–2021*)

- Contributed to coordination and implementation of a comparative survey for university students and adults

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Economics, University of Dhaka (*October 2025 – present*)

Lecturer, Department of Economics, University of Dhaka (*November 2018 – September 2025*)

- Taught 10 undergraduate economics courses; supervised undergraduate dissertations and research projects
- Trained undergraduate and graduate students in STATA; developed computational course materials in R, Python, and STATA

Lecturer, Department of Economics, East West University (*March 2018 – October 2018*)

AWARDS

- Commonwealth Master's Scholarship (2022)
- Gold Medals for highest CGPA, BSS & MSS Economics, University of Dhaka (2016, 2017)

LANGUAGES

Advanced fluency in English and Bangla